PRODUCT PERFORMACE UPDATE

ITC INTERNATIONAL TECHNICAL CERAMICS, LLC

Application: Lehigh Heavy Forge - Forging Furnace

FURNACE DETAILS

- **Lining Configuration:** The forging furnace is approximately 32' L x 20' W x 12' H. The sidewalls are a mix of shotcrete, gunnite and plastic refractory 18" thick. The roof is rammed plastic 9" thick and the back wall and door are 12" thick ceramic fiber module construction.

- **Customer Need:** ITC recommended a solution to address several major challenges,
  - Heat loss and high shell temperatures on the roof and sidewalls due to aged and badly spalled refractory and shrinking and severe devitrification of fiber lining at operating temperatures of $2{,}200^0$ F - $2{,}300^0$ F.
  - Frequent refractory maintenance.

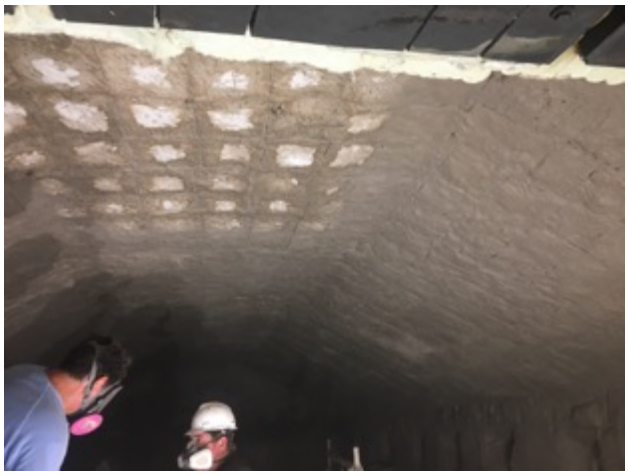
ITC RECOMMENDATION

- **Engineered Approach:** ITC recommended a hot face veneer to address the heat loss through the roof and sidewalls and spray coating the back wall and door to address the shrinking & devitrification of the ceramic fiber.

- **High Temperature Energy Efficient Ceramic Coatings:** ITC manufactures a proprietary, high temperature, energy efficient ceramic coating to work in concert with the furnace conditions and all types of refractory design. Under ideal conditions, high temperature, energy efficient ceramic coatings re-radiate thermal energy from the process to the furnace load, imparting the following benefits…

  Lower heat loss – Minimizes heat transfer through the furnace refractory walls
  Lower Maintenance costs – Protects the substrate refractory
  Improved furnace operation – Improves the thermal efficiency of the process

Existing furnace refractory



Fully veneered and coated furnace

IMPACT ON THE PROCESS

- LOWER SHELL TEMPERATURES
  The high temperature energy efficient ceramic coating that protects the refractory lining, have lowered shell temperature and eliminated hot spots.

- IMPROVED BTU SAVINGS
  Due to the re-radiating properties of the coating, Btu savings are significantly better, average fuel consumption has been reduced by 36% - 40%.

- REDUCED REFRACTORY MAINTENANCE
  Protection from the coating helps eliminate ceramic fiber from shrinking and devitrification and hard refractory from spalling thus reducing the amount of time and material need to keep the furnace operational.

- INCREASED REFRACTORY LONGEVITY
  Once the coating is applied, refractory life is increased by 3 to 5 times its normal life expectancy.

- REDUCED TURN-AROUND TIME
  Average turn-around times are reduced by 60%.